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Physics Internal Assessment

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The influence of sugar concentration on the resonance decay time of a metal cup as measured using digital waveform amplitude data

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Introduction

As a musician, it's the small differences in tonality and resonance that make the difference between an instrument, and a lifelong partner. Acoustics is one of the oldest branches of physics, required in the making of the first instruments nearly 60,000 years ago (Narodni Muzej Slovenije), and has been constantly perfected since, with breakthroughs coming to this day as new observations are constantly made. For example, when striking a metal cup, much like a xylophone, it produces a bell-like note as the vibrations settle at its resonance frequency. By adding water to the cup, we can influence its pitch but interestingly enough, it also affects how long the sound stays as a result of energy loss due to dampening. This second fact caught me by surprise and got me to question whether changing the nature of the liquid in the form of viscosity could influence the duration of dampening for the sound.

Research question

How does the mass concentration of sugar in water (0%, 20%, 40%, 50%, 60%, 80%, 100%) affect the resonance decay time (τ) of sound in a metal cup as measured using digital waveform amplitude data.

Background Information

All objects and systems that can vibrate have an intrinsic property known as natural frequencies. This is defined as the set of frequencies at which the system vibrates naturally when disturbed by an external force. An example of this is a tuning fork, which oscillates at a set frequency, depending on its mass, stiffness, and geometry. Similarly, a metal cup with liquid forms a system that should have its own natural frequency. The frequency f can then be defined as following:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$ (equation 1) (Fiveable)

Where k is the stiffness constant in $\frac{N}{m}$, and m is the vibrating mass in Kg of the system

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Ideally, the system would vibrate at this frequency indefinitely, however energy lost to the surroundings through air resistance and internal friction results in the amplitude of the resultant sound to fade with time. This phenomenon due to the dissipation of mechanical energy is known as damping. In this investigation, the primary source of damping in the system comes from the syrup inside the cup due to the resistive force it implements against the velocity of the surface of the cup due to viscosity.

Because of this significant damping effect, the cup's system can be modeled as a damped harmonic oscillator. The motion of this system can then be approximated to the following second-order differential equation :

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0 \quad \text{(equation 2) (Libretexts)}$$

Where m is the vibrating mass in Kg, b is the damping coefficient in $\frac{Kg}{s}$, k is the stiffness constant of the cup in $\frac{N}{m}$, and x is the displacement of the surface of the cup in m.

As is the case for the cup, an underdamped system is defined as one where oscillations decay gradually. For this case, the solution for x to equation 2 gives us,

$$x(t) = A_0 e^{-\frac{b}{2m}t} cos(\omega t) \quad \text{(equation 3) (Libretexts)}$$

Where A_0 is the initial amplitude, $cos(\omega t)$ represents the oscillatory motion at an angular frequency ω, and $e^{-\frac{b}{2m}t}$ accounts for the exponential decay of amplitude with time due to damping.

$\frac{2m}{b}$ can also be represented as τ. It represents the time it takes for the amplitude to fall by a factor of $\frac{1}{e}$. substituting τ back into equation 3, we can obtain a function for amplitude with respect to time as shown:

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$$A(t) = A_0 e^{-\frac{t}{\tau}} \quad \text{(equation 4)}$$

On any recorded waveform, we can plot 2 points along its amplitude envelope in the form (A_1, t_1), and (A_2, t_2). Then, using equation 4, we can find τ, the aim of our investigation, for each waveform like so:

$$A_1 = A_0 e^{-\frac{t_1}{\tau}}, A_2 = A_0 e^{-\frac{t_2}{\tau}}$$

$$\frac{A_1}{A_2} = \frac{A_0 e^{-\frac{t_1}{\tau}}}{A_0 e^{-\frac{t_2}{\tau}}} = e^{-\frac{t_1}{\tau}} e^{\frac{t_2}{\tau}}$$

$$\frac{A_1}{A_2} = e^{\frac{t_2 - t_1}{\tau}} = e^{\frac{\Delta t}{\tau}}$$

$$ln(\frac{A_1}{A_2}) = \frac{\Delta t}{\tau}$$

$$\tau = \frac{\Delta t}{ln(\frac{A_1}{A_2})} \quad \text{(equation 5)}$$

This equation will be crucial to this investigation as this is how the τ values will be collected.

Viscosity (η) refers to a fluid's internal resistance to flow in $\frac{Pa}{s}$. It results from the friction between microscopic adjacent layers of fluid travelling at different velocities. As you increase sugar concentration in water, sugar molecules form hydrogen bonds with water molecules, hindering molecular motion, and thus increasing the viscosity of the water. Literature shows that η increases exponentially with concentration C in $\frac{mol}{L}$,

$$\eta = 0.95e^{C} - 0.006 \quad \text{(equation 6) (Kim and The University of British Columbia)}$$

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As the cup vibrates, the viscous drag force of the sugar syrup inside resists the oscillating motion, thus increasing the damping coefficient b. Since the damping coefficient is inversely proportional to τ,

$$\tau = \frac{2m}{b} \quad \text{(equation 7)}$$

as,

$$\eta \propto b \quad \text{(equation 8)}$$

and,

$$\tau \propto \frac{1}{b} \quad \text{(equation 9)}$$

We can then find a relationship between τ and η being,

$$\tau \propto \frac{1}{\eta} \quad \text{(equation 10)}$$

Combining with equation 6,

$$\tau \propto \frac{1}{0.95e^{C} - 0.006} \quad \text{(equation 11)}$$

Given that $0.95e^{C} \gg 0.006$, the final relation between τ and C can be given as,

$$\tau \propto e^{-C} \quad \text{(equation 12)}$$

Hypothesis

As the mass-volume concentration of sugar increases, τ will decay exponentially. This is supported by equation 6 showing an exponential increase in η with C, and equation 10 that suggests an inverse relationship between η and τ. Combining, this gives,

Variables

Independent variable

The mass- volume concentration of sugar in the sugar syrup (0%, 20%, 40%, 60%, 80%, 100%): These values were chosen as they cover a broad range of concentration values. It was also found

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through testing that at higher concentrations, the sugar would precipitate from the solution at room temperature.The syrup will be prepared with 500g of sugar to create a $500cm^3$ solution of syrup, corresponding to 100% concentration. As we decrease the concentration, the syrup transferred in the cup will be diluted with water until the desired concentration is reached. The uncertainty in the concentration comes from the apparatus (measuring cylinders and scales) used in creating the solution.

Dependant variable

The resultant τ values: Each time the cup is hit, a microphone will record the audio and create a waveform on audacity. Then using automeris.io, 2 points along the amplitude envelope will be selected, providing 2 points (A_1, t_1), and (A_2, t_2). Using equation 5, these points will be processed to find the τ value for each of the trials. It is assumed that there is no uncertainty as the numerical plots are accurate to a large number of significant figures.

Control variables

***Table 1**: Controlled variables*

Control variable	Implication	Method of control
Microphone	Differences in calibration can have an effect on the recorded relative amplitude, despite the same sound.	The same laptop microphone will be used in each run
Microphone distance	Acoustic attenuation dictates that the distance between the microphone and cup will influence the recorded sound intensity.	The computer microphone is always $10cm$ away from the edge of the cup
The temperature of the solution	As the temperature changes, so does the energy of the molecules to overcome attractive forces, influencing viscosity.	Using a thermometer, the temperature is verified to be $20°C \pm 0.1$

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Recording software	Differences in in-app processing and data handling may have an effect on the wave's resolution	All runs will be recorded on Audacity
The cup used	Even if the same model, small differences in shape can influence the resonance frequency, and thus dampening behavior	The same cup is rinsed and used in each run
Table material	Depending on the material, mechanical energy may dissipate differently from the cup system.	The runs are conducted on the same table, on the same area
Volume of solution	The volume of solution affects its surface of contact with the cup, affecting the rate of energy dissipation	Each run will have $150ml$ of solution in the cup

***Table 2**: Uncontrolled variables*

Uncontrolled Variable	Implication	Why is it not?
Atmospheric conditions	Humidity and temperature of the surrounding air may affect the emitted sound waves recorded	No access to an AC or other climate control systems
Disturbance force administered	The initial force with which the cup is hit will determine the initial amplitude, and itself disturbs the system	Human error can not be avoided without expensive machinery, but the mathematical processing theoretically does not depend on it.

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Apparatus and Materials

Table 3: Apparatus and materials

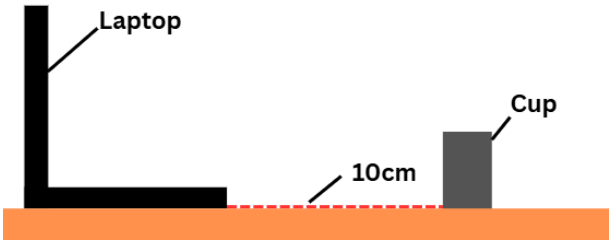
Apparatus	Materials
1 x $500cm^3$ Measuring Cylinder ($\pm$ $0.1cm^3$) 1 x Weighing scale (accurate to $0.1g$) 1 x $20cm$ ruler ($\pm$ $0.1cm$) 1 x metal cup 1 x Computer with a microphone and audacity preinstalled	White sugar Distilled water

Methodology

Procedure development

When designing the experiment, it was important to ensure that as many factors were kept constant across the trials as possible. By considering the controlled variables above, the following experiment setup was devised:

Figure 1: Experiment setup



Here, the cup used is the same across runs, and so is its position relative to the computer. The experiment is conducted on the same area of the table to ensure the honeycomb interior does not cause different rates of energy dissipation.

We can also discuss the implications of some of the uncertainties in this experiment. The concentration of sugar will have an uncertainty due to the weighing scale and the measuring cylinder. We can calculate it as follows:

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$$C = \frac{m_{sugar}}{v_{solution}},$$

$$\text{thus, } \frac{\Delta C}{C} = \sqrt{\left(\frac{\Delta m_{sugar}}{m_{sugar}}\right)^2 + \left(\frac{\Delta V_{solution}}{\Delta V_{solution}}\right)^2} = \sqrt{\left(\frac{0.1g}{500g}\right)^2 + \left(\frac{0.1ml}{500ml}\right)^2} = 0.000283$$

Converting to percentage uncertainty gives, $0.000283 \times 100 = 0.0283\% \approx 0.03\%$

This value is then adjusted for the different solution dilutions.

Although relatively small, it is important to note its potential influence on the final conclusions drawn as we have established its direct influence on the dependent variable.

We must also consider the uncertainty of the microphone. The specifics of my computer microphone were not available, but we must note that the amplitudes recorded are subject to the specific model's uncertainty. This will have a direct influence on the points extrapolated from the recordings, and subsequent τ calculations.

Procedure

1. Preparation on the sugar solution:
    a. Using the weighing scale, measure 500 grams of sugar and transfer it into the measuring cylinder.
    b. Continue to transfer distilled water into the cylinder until a volume of 500ml is achieved (continuously mix the sugar to release any air bubbles that may be present)
    c. Continue to stir the contents until the sugar is fully dissolved (note that heat may be used to accelerate the process, however minimal caramelisation must be ensured and evaporated water must be compensated for.
    d. The remaining solution will be a 100% concentration by mass sugar syrup. It will be used in subsequent steps.
2. Recording sound waves
    a. Open Audacity on the computer and turn on the microphone

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   b. Using the ruler, mark a point 10cm away from the microphone and place the cup on it
   c. Pour 150 milliliters of distilled water (or the specific solutions in later trials) into the cup
   d. Begin recording on the computer, then strike the cup hard with a pencil
   e. Wait for the sound to completely dissipate before pausing the recording
   f. Repeat three times for the same concentration to minimise random error and ensure reliability
   g. Using the sugar syrup, prepare 20, 40, 60, 80, and 100% concentration by mass solutions
   h. Repeat steps c-f for each of the concentrations, there should be 18 total sound waves
3. Collecting data points
   a. Using external software such as automeris.io, plot 2 points along the amplitude envelope for each of the sound waves
   b. On a table, record the coordinates for each of the points next to the corresponding run

Risk assessment

Safety wise, It is imperative to be cautious when handling conductive liquids around electronic devices as they can result in electric shocks, or potential shortcircuitting. Spills should also be prevented as sugar syrup can attract pests, and lead to microbacterial infections. To prevent these, the experiment will be carried out in a clean environment, and should not be rushed to prevent hasty mistakes

Environmentally, the materials used at their respective concentration are safe to be disposed down the drain, however that does not change having to be conservative with the quantities used as an excess of any external chemical in an environment will have effects on the ecosystem. Furthermore, disposing of extra syrup was done after dilution to prevent clogging, or pest attraction. Sugar syrup created in the lab should not be used for cooking applications as toxic contaminants may be present.

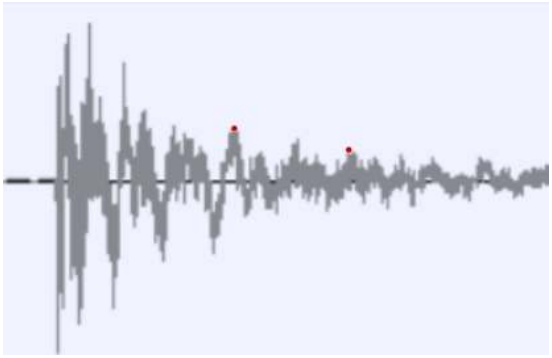
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As there were no living organisms used, there are no outstanding ethical concerns other than those outlined by the environmental concerns of this investigation.

Raw Data

From the experiment, we obtain multiple sound wave profiles, one for each run. As per step 3, we can plot 2 points along the envelope of the wave and extract 2 unique points for further calculations. An example wave is shown below:

***Figure 2**: Sample wave from trial 1, 80% concentration with plotted points*



For each of the trials, we can record the co-ordinates of these points to get the following table

***Table 3**: Raw data from experiment (extrapolated points)*

Concentration of sugar by mass (%)	Trial number	Time of point 1 (t_1) in s to 6 $d.p.$	Amplitude of point 1 (A_1) to 8 $d.p.$	Time of point 2 (t_2) in s to 6 $d.p.$	Amplitude of point 2 (A_2) to 8 $d.p.$
0	1	3.9882857	0.45000000	4.019143	0.25000000
	2	9.5082142	0.41666667	9.550000	0.18750000
	3	12.233541	0.64625000	12.282326	0.25000000
20 ± 0.01	1	3.898165	0.50000000	9.939450	0.20312500
	2	6.175598	0.32378943	6.203576	0.17168977

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	3	10.823282	0.45594595	10.857252	0.21621622
40 ± 0.02	1	4.071884	0.71428571	4.118693	0.21428571
	2	6.355785	0.65517241	6.401240	0.20413793
	3	8.519672	0.39333333	8.528279	0.31666667
60 ± 0.02	1	3.100699	0.37390527	3.127131	0.15951469
	2	5.294525	0.36322002	5.316747	0.17310789
	3	7.504637	0.60603353	7.554799	0.12195000
80 ± 0.03	1	1.694871	0.77941176	1.721762	0.27941176
	2	3.97971	0.78787879	4.008056	0.27272727
	3	6.305856	0.78125000	6.333333	0.26562500
100 ± 0.03	1	4.371184	-0.43857140	4.392129	-0.16142857
	2	6.355743	-0.36111111	6.375000	-0.13888889
	3	8.549327	-0.60000000	8.572480	-0.18571429

Data Processing

Calculating τ

As demonstrated earlier, τ can be calculated using equation 5:

$$\tau = \frac{\Delta t}{ln(\frac{A_1}{A_2})}$$

A sample calculation using Trial 1 at 0% concentration would go as follows:

$$\tau = \frac{(4.019143 - 3.9882857)}{ln(0.45/0.25)}$$

$$\tau = \frac{0.03085724}{ln(1.8)}$$

$$\tau \approx 0.052497 \text{ (accurate to 6 } d.p.\text{)}$$

Repeating this process for every run gives us:

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Table 4: τ values for each run

τ at each run is s		Concentration of sugar by mass (%)					
		0	20 ± 0.01	40 ± 0.02	60 ± 0.02	80 ± 0.03	100 ± 0.03
Run	1	0.052497	0.045832	0.038878	0.031029	0.026213	0.020957
	2	0.052330	0.044102	0.038980	0.029986	0.026719	0.020153
	3	0.051367	0.045530	0.039697	0.031286	0.025470	0.019743

Calculating mean τ

The mean τ at any given concentration can be given as:

$$\bar{\tau} = \frac{\tau_1+\tau_2+\tau_3}{3}$$

At 0% concentration, this gives us:

$$\bar{\tau} = \frac{0.052497+0.052330+0.051367}{3} \approx 0.052065$$

The uncertainty of the average of 3 values is given by:

$$\Delta\tau = \frac{\sqrt{\frac{1}{3}\sum_{i=1}^{3}(\tau_i-\bar{\tau})^2}}{\sqrt{3}}$$

Which for 0% concentration gives us:

$\Delta\tau = 0.000288 \approx 0.0003$ (we round the uncertainty up to the highest significant figure)

Repeating for all concentrations gives us the final processed data:

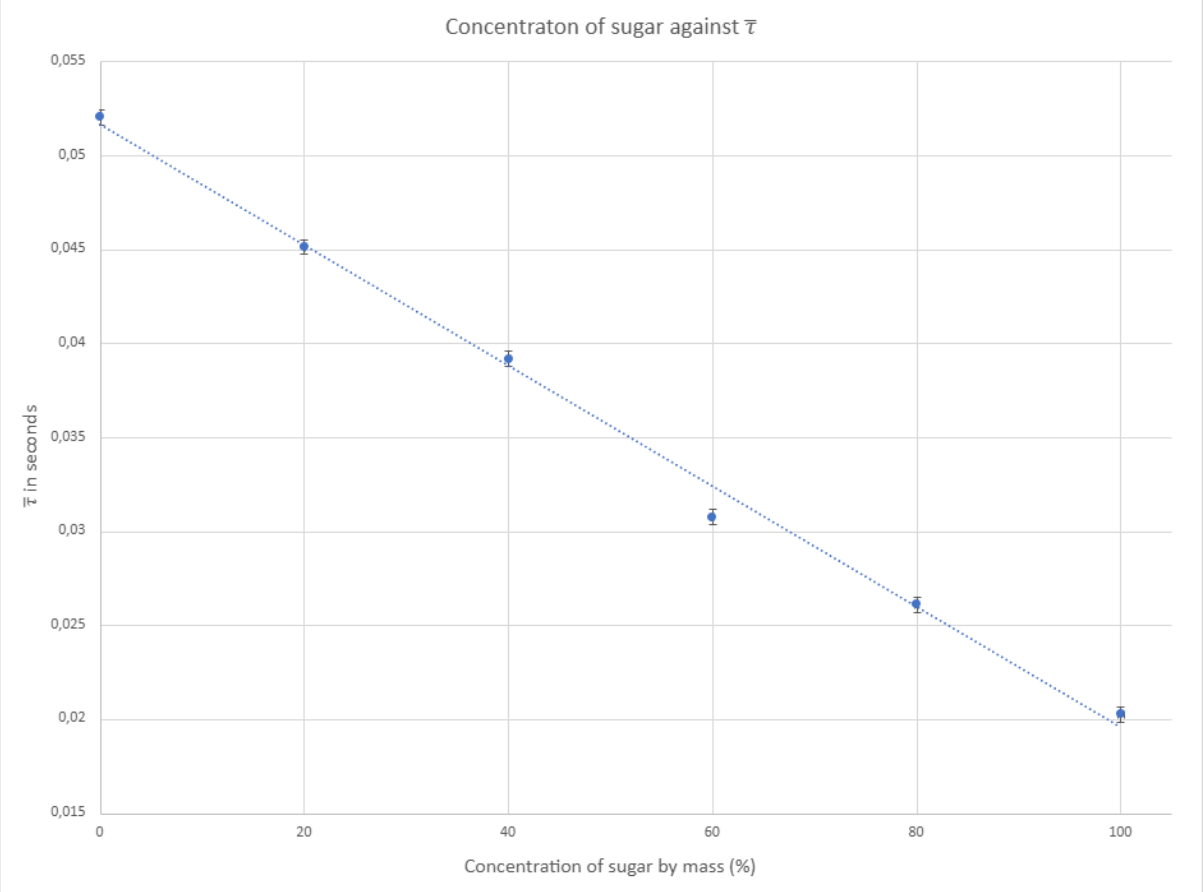
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Table 5: Average τ value at each concentration

Concentration of sugar by mass (%)	$\bar{\tau}$ in seconds
0	0.05206 ± 0.0003
20 ± 0.01	0.04515 ± 0.0004
40 ± 0.02	0.03919 ± 0.0003
60 ± 0.02	0.03077 ± 0.0004
80 ± 0.03	0.02613 ± 0.0003
100 ± 0.03	0.02028 ± 0.0003

This can represented graphically as:

Figure 3: Graph of average τ value at each concentration



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The graph seems to demonstrate a negative linear proportionality between the concentration and $\bar{\tau}$. As such, the line of best fit is also a linear function with the equation $\bar{\tau} = 0.0003C + 0.0516$. The error bars for the point at 60% concentration, however, do not fit on the line, indicating either faulty data collection due to experimental errors, or that this is not the correct relationship. However, the r squared value provided by excel is 0.9952, indicating that this linear relationship is strong

It is interesting to note that the equation implies that there exists points where with a high enough concentration, the time taken for the sound to dissipate is less than 0 seconds. It is important to note that due to the various systematic and random errors, and uncertainties discussed earlier, the real trend may not be reflected by these results.

Data Interpretation and Conclusion

The aim of this investigation was to determine the relationship between the mass-volume concentration of sugar syrup in a cup, and its resonance decay time. As expected, τ decreased with an increase in C. Deviating from my hypothesis, however, is the type of correlation. As per equation 12,

$$\tau \propto e^{-C} \text{ (equation 12)}$$

we should be expecting an exponential decay but the one found was linear. There are several reasons why this discrepancy may be. Firstly, the equations used were idealisation of real world systems. The damping coefficient is not purely dependent on viscosity, but on factors such as energy transfers to the table, internal structural losses of the cup, and acoustic radiation to the air. As they are largely independent of viscosity, the influence of the concentration is therefore reduced on the final τ value.

This could suggest that the other factors of damping can mask the influence of theoretically non-linear factors in these limited concentration ranges, making linear approximations more appropriate to describe observed motion.

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That being said, it must be considered that these results are not precise. This methodology was subject to both systematic, and random errors, as well as uncertainties that influenced the raw data, the processing, and consequently the conclusions drawn as discussed earlier, and in the evaluation.

Outside of the lab, this investigation demonstrates how theoretical systems can deviate from expected theory in real world situations. In the case of dampening, other energy loss systems must be considered and how they might interact with what is being measured. It is thus imperative to perform product testing to uncover real relationships.

Evaluation

***Table 6**: Limitations of the experiment*

Limitation	Significance	Possible improvement
Multiple damping mechanisms present	As discussed, the measured decay constant represents combined forms of energy loss. This reduces the dominance of viscosity and can suppress the expected exponential dependence of τ on concentration.	Suspend the cup using elastic supports or foam isolation to minimise external energy losses and increase the relative contribution of viscous damping.
Variable excitation force	Manual striking introduces variation in the initial impulse and excites different vibrational modes, which can distort the initial decay envelope used to extract τ.	Use a simple mechanical striker to ensure a consistent impulse across all trials.
Uncalibrated microphone response	Unknown microphone sensitivity and frequency response can bias amplitude measurements, particularly at low amplitudes near the noise floor.	Use a calibrated external microphone with fixed gain settings and verified response.
Limited damping regime explored	Over the investigated concentration range, viscous damping may not dominate strongly enough for exponential behaviour to emerge clearly, causing the inverse exponential relationship to appear linear.	Extend the concentration range further by controlled heating to maintain solubility, allowing access to a regime where viscous damping dominates.

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